

Adaptive Learning Platform for Dyslexic Child

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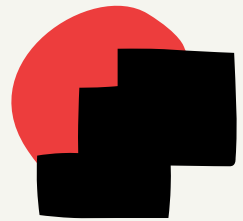
Our Ideas

Open Innovation (AI-Agent + Blockchain)

Introduction:- Our idea focuses on developing an Adaptive Learning Platform specifically designed for dyslexic children, who often encounter difficulties in reading and listening. Dyslexia can lead to challenges such as slow reading skills, confusion with similar letters (e.g., confusing 'b' and 'd'), and issues with writing accuracy.



The Problem



Dyslexic children struggle with conventional learning methods, which can hinder their educational progress and self-confidence. Traditional approaches fail to accommodate their unique learning needs, leading to frustration and a lack of motivation.

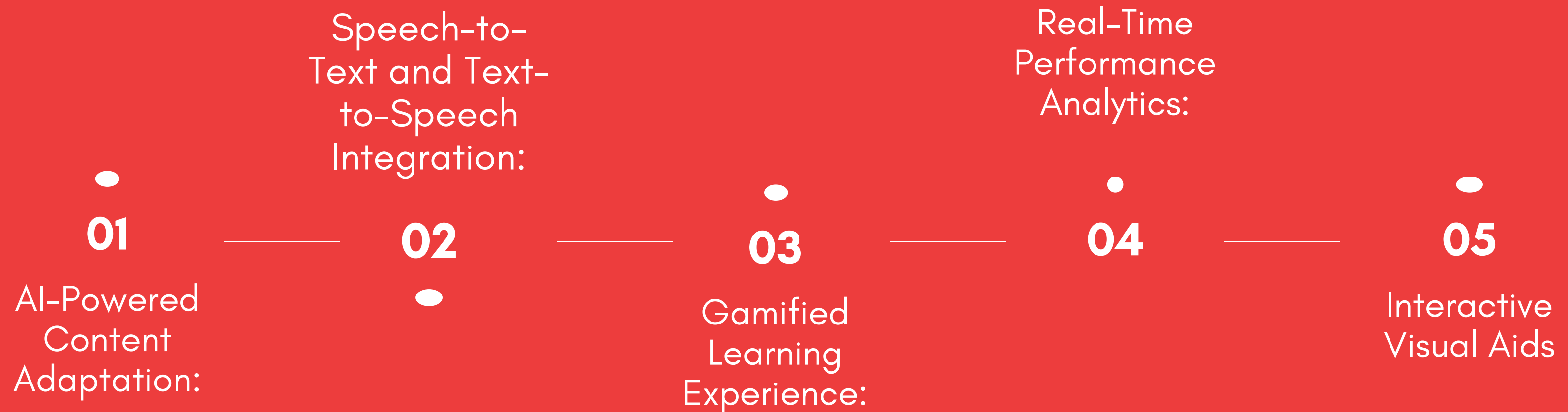


Proposed Solution

We propose an AI-powered Adaptive Learning Platform that leverages Artificial Intelligence (AI) and Blockchain technology to provide a personalized, secure, and engaging learning experience for dyslexic children.



Key Features





Impact and Benifites

Increased Confidence: Children can learn at their own pace without external pressure.

Parent and Educator Collaboration: Provides insights to customize teaching approaches.

Scalability: Can be implemented across various educational institutions globally.

Enhanced Learning Experience: Tailored content ensures better comprehension.



Terms and Technologies Used

- **Artificial Intelligence (AI):** Personalizing content and tracking learning progress.
- **Blockchain Technology:** Ensuring secure and transparent data management.
- **Speech-to-Text Technology:** Converting spoken words into written text.
- **Text-to-Speech Technology:** Converting written content into audio.
- **Gamification:** Using game elements to enhance engagement and motivation.
- **Performance Analytics:** Generating reports for insights and improvement tracking.



APIs Used



- **Mozilla DeepSpeech API:** An open-source speech-to-text engine for transcribing spoken words into text.
- **eSpeak API:** A free text-to-speech synthesizer supporting multiple languages.
- **CMU Sphinx API:** An open-source speech recognition system for voice input processing.
- **OpenAI Whisper API (free tier):** Transcription and speech recognition capabilities.
- **TensorFlow.js:** A free library for machine learning in web applications.
- **Firebase API (Free Tier):** Real-time database for storing and retrieving learning progress.
- **Blockchain API (Hyperledger Fabric – Community Edition):** Open-source solution for secure student data management.



Conclusion



Conclusion Our Adaptive Learning Platform empowers dyslexic children by offering an inclusive and personalized educational experience, leveraging AI and blockchain to address their unique learning challenges and help them achieve their full potential.

